

Comparison of Homogenization Time in Mosquito DNA Extraction

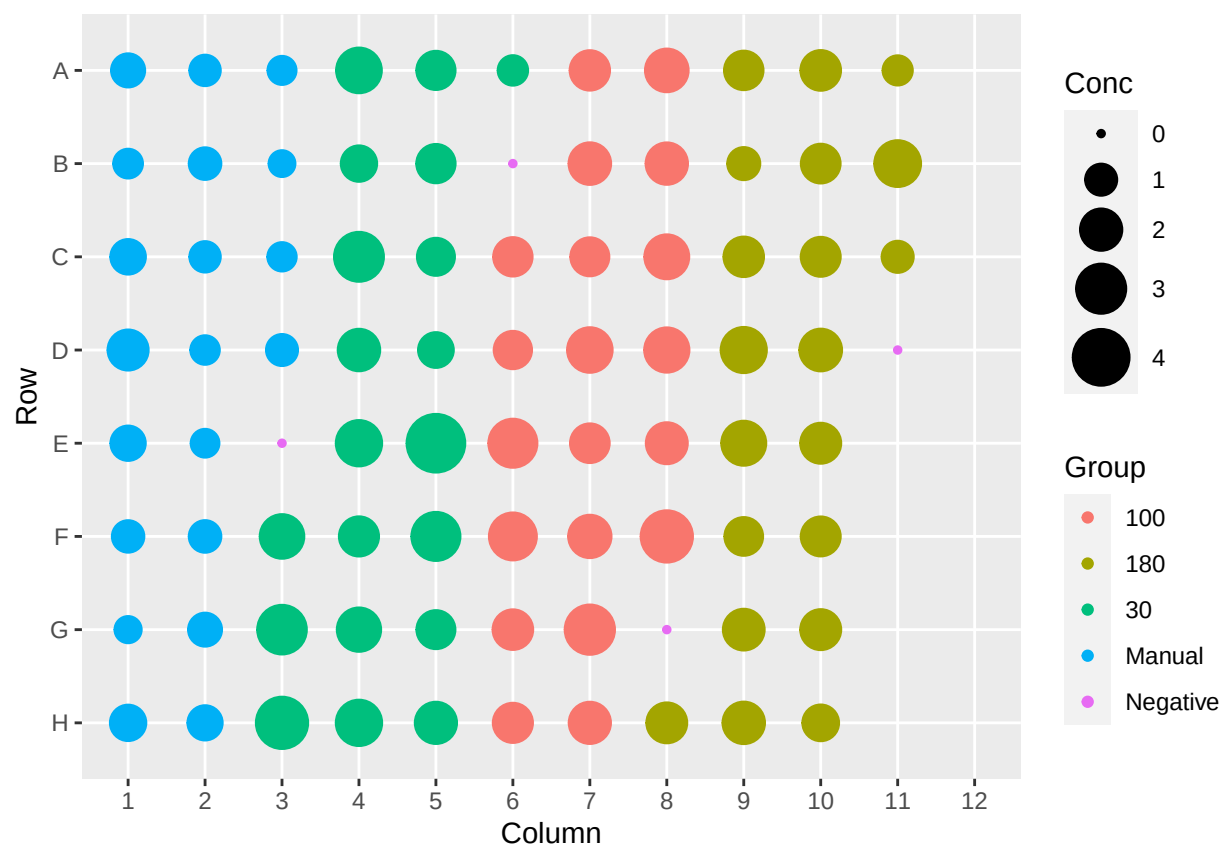
Jack Dorman

2/18/2020

The purpose of these runs was to identify the ideal time to homogenize samples for in the TissueLyser prior to extraction using the QIAcube HT. Insectary raised *Anopheles gambiae* from Johns Hopkins University were used. These mosquitoes were infected with *Plasmodium falciparum*. Extraction followed the Isolation of Genomic DNA from Mosquito Thoraxes (QIACUBE HT) SOP which can be accessed from the “protocols” folder of Data Depot.

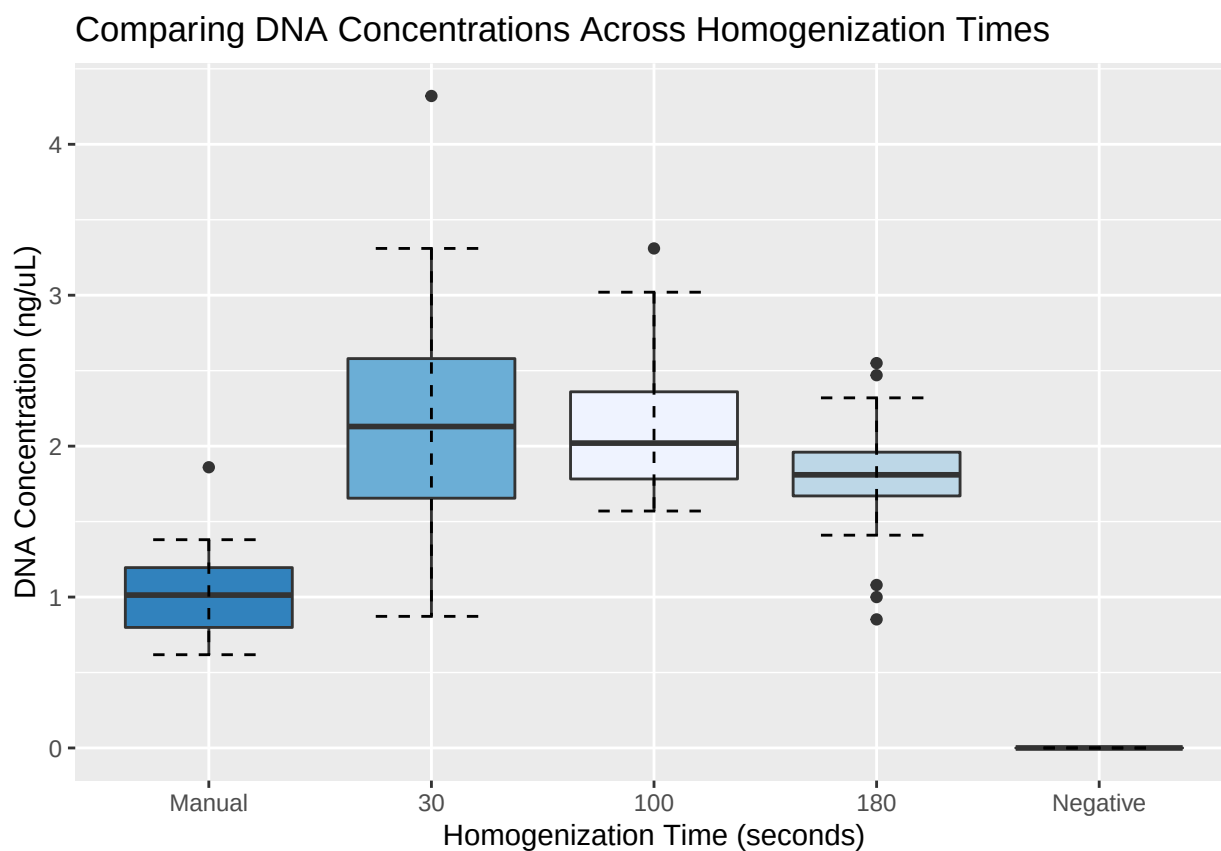
Plate layout for Thorax Run

Color corresponds to treatment group while size correlates with the approximate DNA concentration (ng/uL).



Analysis of Thorax gDNA Extract Concentration

DNA concentration was assessed with HS Qubit 1x dsDNA



Trends in Concentration by Plate Column

This is a quality control measure to assess any potential biases that could result from a compromised run.

Comparing DNA Concentrations Across Columns

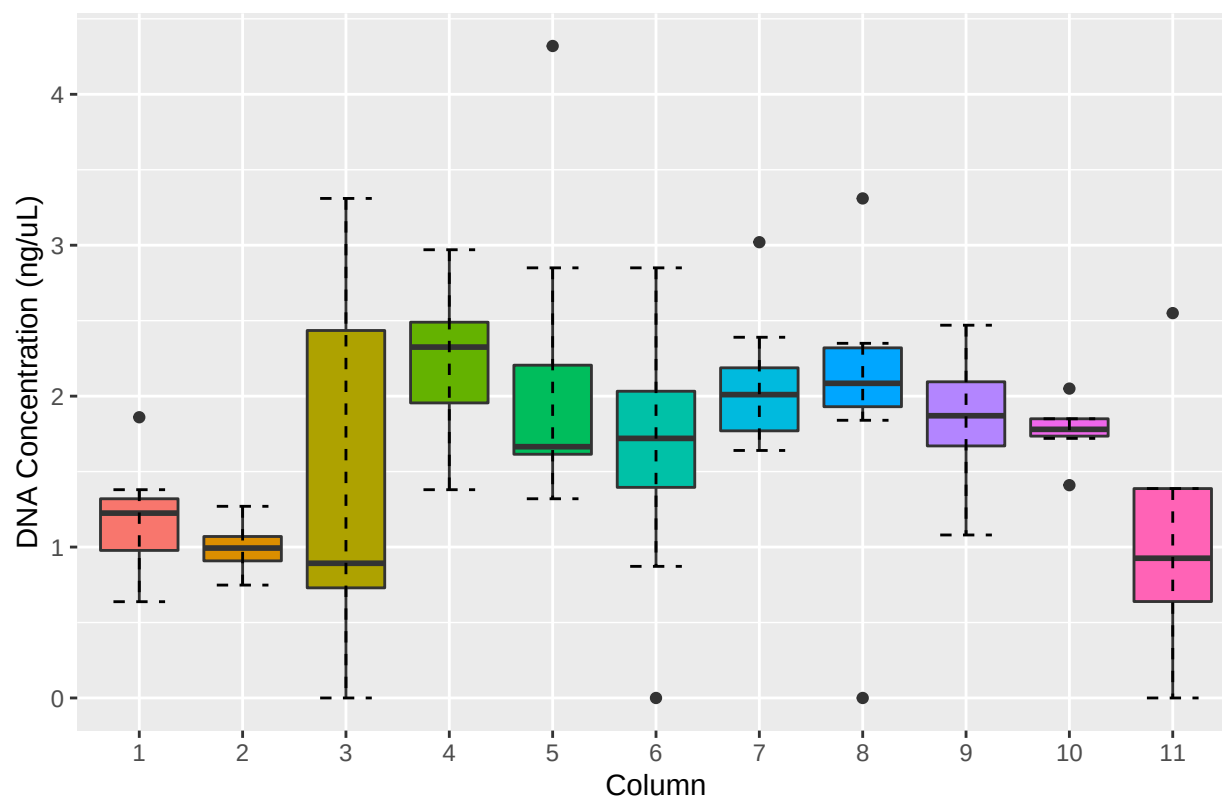
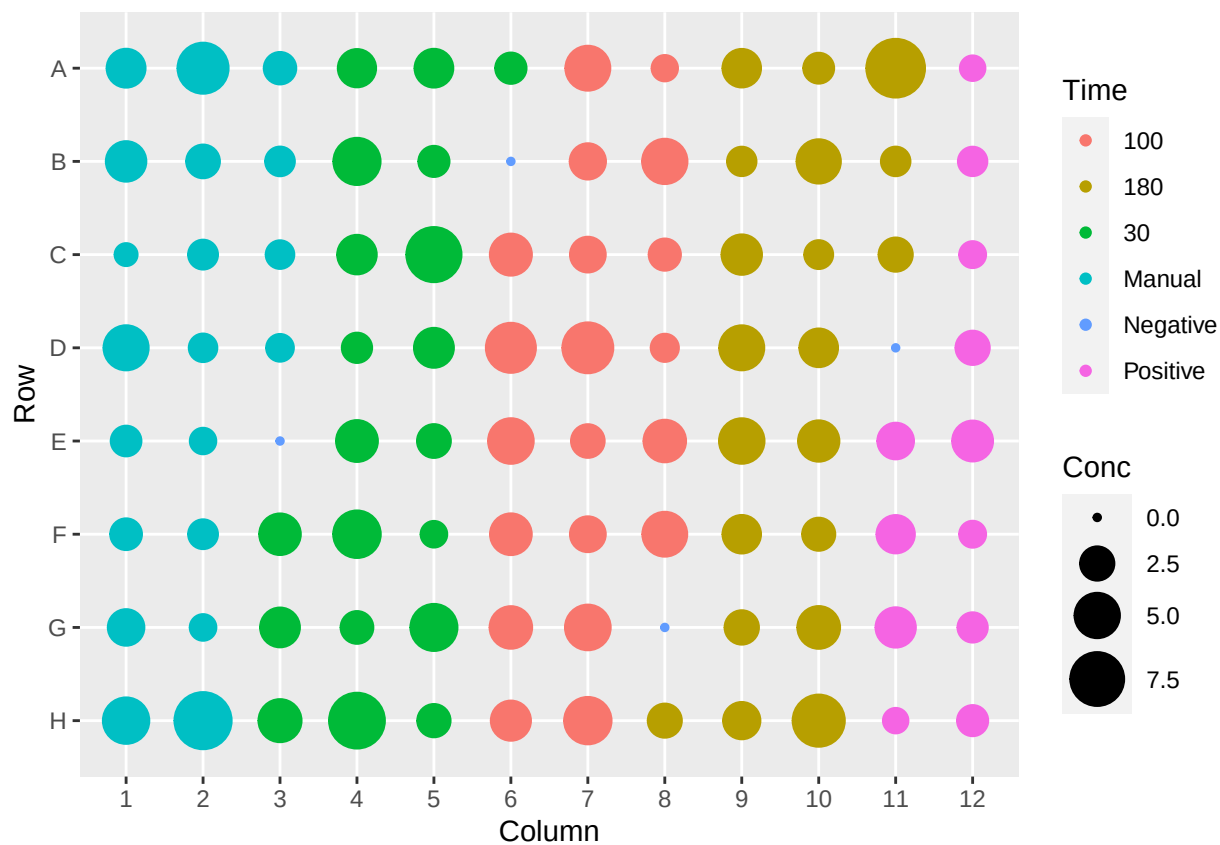


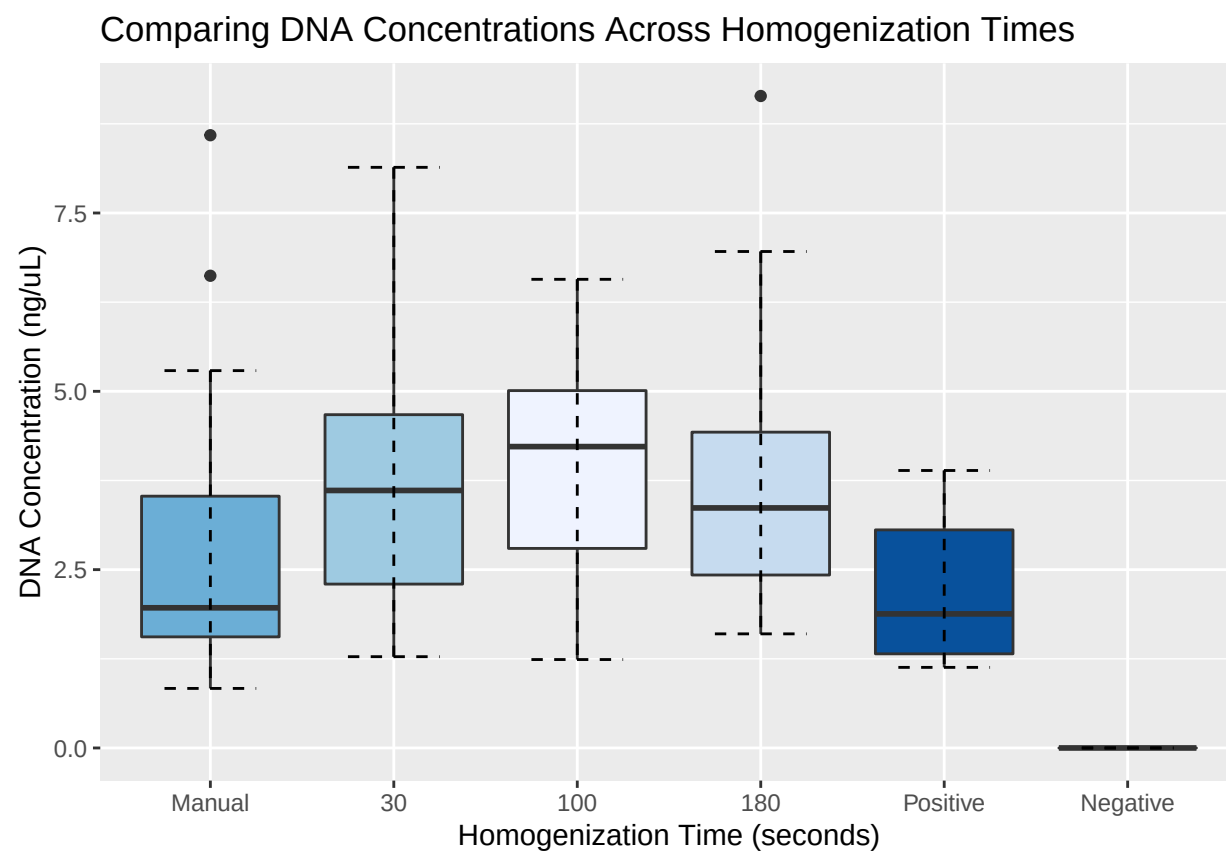
Plate Layout for Abdomen Run

Color corresponds to treatment group while size correlates with the approximate DNA concentration (ng/uL). The Positive group corresponds to thoraxes homogenized for 100 seconds (the best performing time from the previous run).



Analysis of Abdomen gDNA Extract Concentration

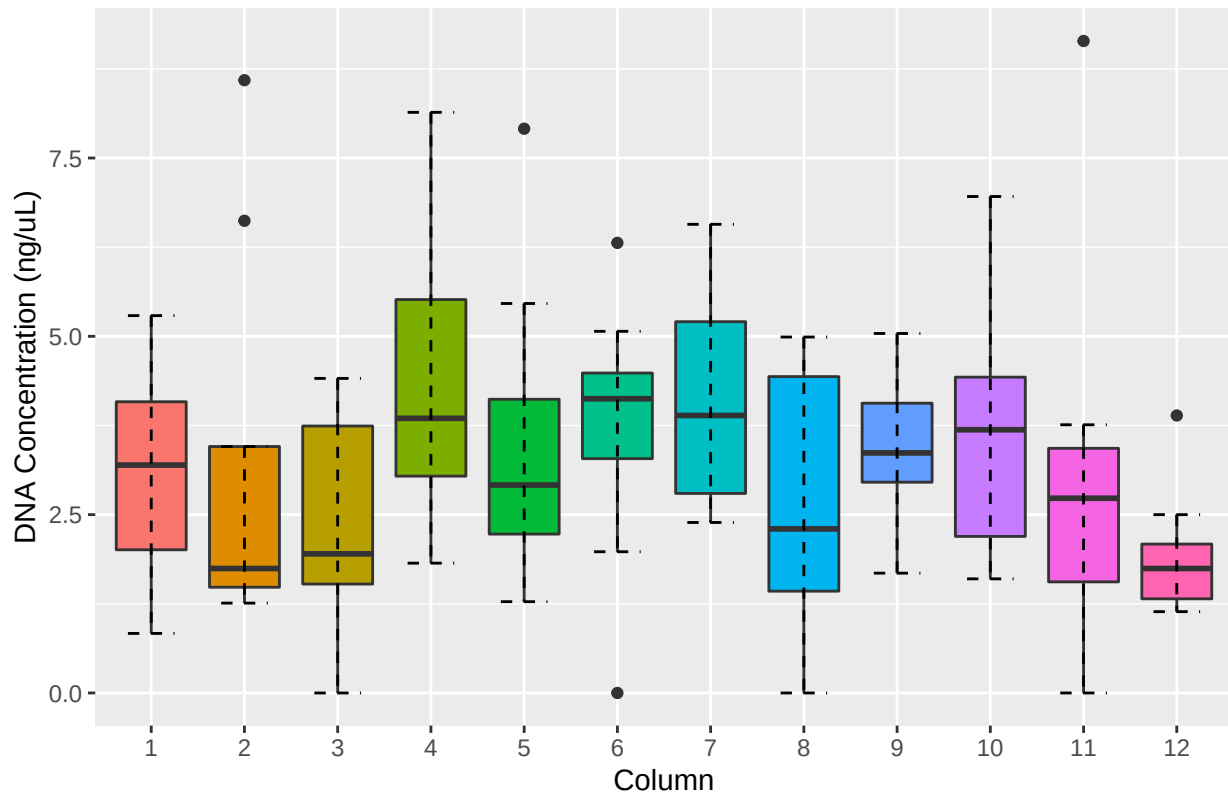
DNA concentration was assessed with HS Qubit 1x dsDNA



Trends in Concentration by Plate Column

This is a quality control measure to assess any potential biases that could result from a compromised run.

Comparing DNA Concentrations Across Columns



Summary

Overall, 100 seconds seems ideal for thoraxes. It is the first group that demonstrates a statistically significant improvement in DNA yield relative to the manual homogenization group. TapeStation corroborates this assessment with fragmentation not significantly increasing between the 30 and 100 second groups (see Feb_MosExt_Run1_Report under TapeStation_Data in data of Data Depot). For abdomens the trends are less clear with no group demonstrating statistically improved DNA yield as compared to manual homogenization. The 100 second group demonstrated the highest average however it was within the range of the 30 second group. TapeStation data seemed to indicate 30 seconds produced the highest quality peaks compared to any group but it was not significantly improved over 30 seconds (see MosDNAExt_Abdomen_Homog_021920 in the same location as previously mentioned). All quality control measures (column comparison, positive, and negative controls) indicated the runs were successful.